[¿=Latex, font=, tier$_c ol/.style = draw = 1!70!black, fill = 1!2!white, rounded corners = 6pt, line width = 0.7pt, col_t$
[tier$_c ol = blue, minimum width = 5.2cm, minimum height = 5.8cm](col1) at (2.8, 0); [col$_t$itle = blue, minimum width
[flow$_i$nput = blue](t1$_i$n) at (2.8, 2.0) **Input: Full Evidence Graph** G_q Unpruned multigraph with guided paths caffold;
[flow$_p$roc = purple](t1$_l$lm) at (2.8, 0.95) **Primary LLM Reasoning (Call 1)** *Strict audit*• Linear path evaluation ;
[flow$_d$ec = purple](t1$_d$ec) at (2.0, −0.2) Unambiguous Evidence?;
[flow$_e$xit, minimum width = 1.35cm](t1$_e$xit) at (4.45, −0.2) **Fast Exit**≈ 80% Queries
1 LLM Call ;
[flow$_p$roc = orange, fill = orange!5, draw = orange!80!black](t1$_h$and off) at (2.8, −1.85) **Diagnostic Handoff** $\mathcal{D}$Collis
[flow$_a$rr](t1$_i$n) − −(t1$_l$lm); [flow$_a$rr](t1$_l$lm) − −(t1$_d$ec); [flow$_s$ucc](t1$_d$ec.east) − −(t1$_e$xit.west); [font =, text = gre
[bridge$_b$lock](bridge) at (6.5, −0.9) **Graph Compression** blue!90!black$_{\text{Zero LLM Calls}}$Fetch$_{\text{Doc:Blues}}$Prune non − tied row steal!90!b
[tier$_c ol = teal, minimum width = 5.2cm, minimum height = 5.8cm](col2) at (10.2, 0); [col$_t$itle = teal, minimum width
[flow$_i$nput = teal](t2$_i$n) at (10.2, 2.0) **Input: Minimal State Capsule** $\mathcal{S}_{min}$ teal!90!black$_{\text{Compressed graph:}}$ Tied candidates + bri
[flow$_p$roc = purple](t2$_l$lm) at (10.2, 0.95) **Targeted LLM Resolution (Call 2)** *Isolated context*• Zero context dilu
[flow$_d$ec = teal](t2$_d$ec) at (10.2, −0.2) Grounded in Bridge Evidence?;
[flow$_t$erm](t2$_t$erm) at (10.2, −1.85) **Final Verified Answer: "York"** ≤ 2 LLM Calls Total • 100% Grounded Provenance ;
[flow$_a$rr, draw = teal!80!black](t2$_i$n) − −(t2$_l$lm); [flow$_a$rr](t2$_l$lm) − −(t2$_d$ec); [flow$_s$ucc](t2$_d$ec.south) − −(t2$_t$erm.n
[flow$_w$ire, rounded corners = 3pt](t1$_h$and off.east) − −(6.5, −1.85) − −(bridge.south); [flow$_a$rr, draw = teal!80!blac